

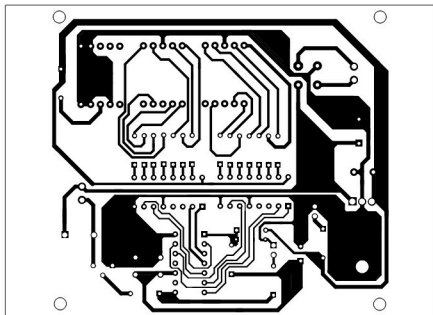


Fig. 3: An actual-size, single-side PCB layout for the MSP430G2231-based temperature indicator and controller

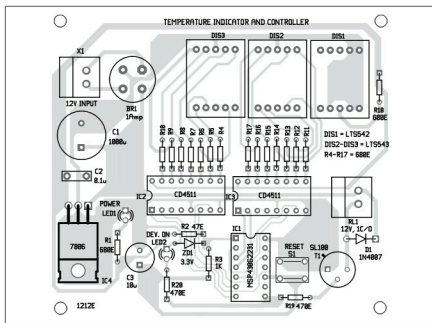


Fig. 4: Component layout for the PCB

explained above in Step 6 under operation of CCS section.

Note that CCS can automatically save the modified source files. Once the code is compiled successfully as mentioned in the step above, you can load the code into the MCU using the LaunchPad board.

Open 'Debug' from 'Run' menu, connect the board and download the code to the target (flash device).

Click 'Debug' button (green bug). The 'Debug' icon in the upper right-hand corner indicates that you are now in 'Debug Perspective' view. If download is successful, carefully pull up the chip with forceps from the board and place it in your circuit.

Temperature conversion. The conversion and calculation are mainly based on the ADC resolution and reference voltage. The microcontroller used

here has a 10-bit ADC, which means there are total 1024 divisions for the entire span.

The MSP430G2231 contains a temperature-sensitive resistor (thermistor) connected to the ADC. The thermistor changes resistance with temperature, changing the voltage input to the ADC. The ADC compares the variable voltage with the reference voltage. The 10-bit ADC on the MSP430G2231 returns a number between 0 and 1023. For example, with an analogue input of 1 volt and the LaunchPad's reference voltage of 1.5 volts, the ADC returns 682 because $1/1.5 = 0.666 = 682/1024$. The same method is applied in the project.

Construction and testing

An actual-size, single-side PCB layout of the MSP430G2231-based temperature indicator and controller is shown in Fig. 3 and its component layout in Fig. 4.

The ambient temperature is captured by the inbuilt temperature sensor of MSP430G2231. When the ambient temperature reaches 30 degrees, the cooler connected across relay RL1 is switched on automatically and the displays shows 30°C. If you want to change the temperature setting, change the value '29' in the code 'if(InCelsius>29)' to the value as per your requirement:

```
{
P2OUT = (BIT7);
P1OUT |= BIT4;
}
if(InCelsius>29)
P1OUT |= BIT6;
else
{
P1OUT &= (~BIT6);
}
}
```

For example, if you want the cooler to turn on at 25 degrees, change the value in the code to '24'. Compile it and burn the code into the MCU using the LaunchPad. Next time, your appliance will turn on when the temperature reaches 25 degrees. ●

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